

Program 7: Deploy a web application to kubernetes using minikube

File structure:

Nginx-deployment.yaml

apiVersion: apps/v1

kind: Deployment

metadata:

 name: nginx-deployment

spec:

 replicas: 2

 selector:

 matchLabels:

 app: nginx

 template:

 metadata:

 labels:

 app: nginx

 spec:

 containers:

 - name: nginx

 image: nginx:latest

 ports:

 - containerPort: 80

Start the minikube service and check the status

 #minikube start

 #minikube status

Create the deployment

 #kubectl apply -f nginx-deployment.yaml

Check the deployment status

 #kubectl get deployments

 #kubectl get pods

Expose the port

 #kubectl expose deployment nginx-deployment --type=NodePort --port=80

Get the service name

 #kubectl get services

```

1RV24MC036_gayatri_v_k@ubuntu:~$ minikube start --driver=docker
🐳 minikube v1.37.0 on Ubuntu 24.04
👉 Using the docker driver based on existing profile
🔥 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
❗ Failing to connect to https://registry.k8s.io/ from inside the minikube container
👉 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
🔄 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
👉 Enabled addons: default-storageclass, storage.provisioner
🏠 Done! kubectrl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC036_gayatri_v_k@ubuntu:~$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

1RV24MC036_gayatri_v_k@ubuntu:~$ mkdir program7
1RV24MC036_gayatri_v_k@ubuntu:~$ cd program7
1RV24MC036_gayatri_v_k@ubuntu:~/program7$ nano nginx-deployment.yaml
1RV24MC036_gayatri_v_k@ubuntu:~/program7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
1RV24MC036_gayatri_v_k@ubuntu:~/program7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment 1/2      2             1           9s
1RV24MC036_gayatri_v_k@ubuntu:~/program7$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-9x2qv  1/1     Running   0           23s
nginx-deployment-6f9664446b-bqrs6  1/1     Running   0           23s
1RV24MC036_gayatri_v_k@ubuntu:~/program7$ kubectl expose deployment nginx-deployment \
--type=NodePort \
--port=80
service/nginx-deployment exposed
1RV24MC036_gayatri_v_k@ubuntu:~/program7$ kubectl get services
NAME          TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes    ClusterIP     10.96.0.1     <none>         443/TCP           11h
nginx-deployment  NodePort      10.107.197.115 <none>        80:32327/TCP      8s
1RV24MC036_gayatri_v_k@ubuntu:~/program7$
nginx-deployment NodePort <cluster-ip> <none> 80:31xxx/TCP
bash: syntax error near unexpected token '<'
1RV24MC036_gayatri_v_k@ubuntu:~/program7$ nginx-deployment NodePort <cluster-ip> <none> 80:31xxx/TCP

```

```

1RV24MC036_gayatri_v_k@ubuntu:~/program7$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-9x2qv  1/1     Running   0           23s
nginx-deployment-6f9664446b-bqrs6  1/1     Running   0           23s
1RV24MC036_gayatri_v_k@ubuntu:~/program7$ kubectl expose deployment nginx-deployment \
--type=NodePort \
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service/nginx-deployment exposed
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1RV24MC036_gayatri_v_k@ubuntu:~/program7$ minikube service nginx-deployment

```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:32327

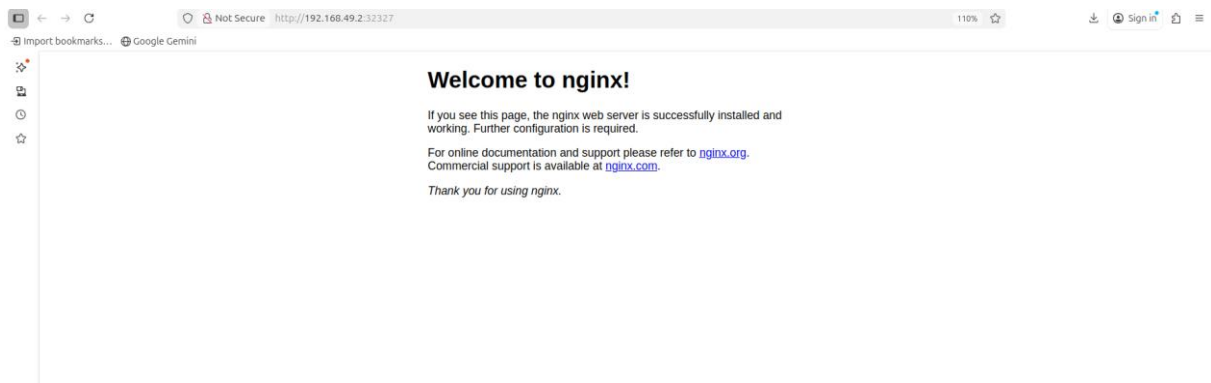
```

🔗 Opening service default/nginx-deployment in default browser...
1RV24MC036_gayatri_v_k@ubuntu:~/program7$ Gtk-Message: 09:31:08.339: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
1RV24MC036_gayatri_v_k@ubuntu:~/program7$

```

Start the service

#minikube service nginx-deployment



Deleting resources

```
#kubectl delete pods --all
```

```
#kubectl delete svc--all
```

```
#kubectl delete deployments --all
```